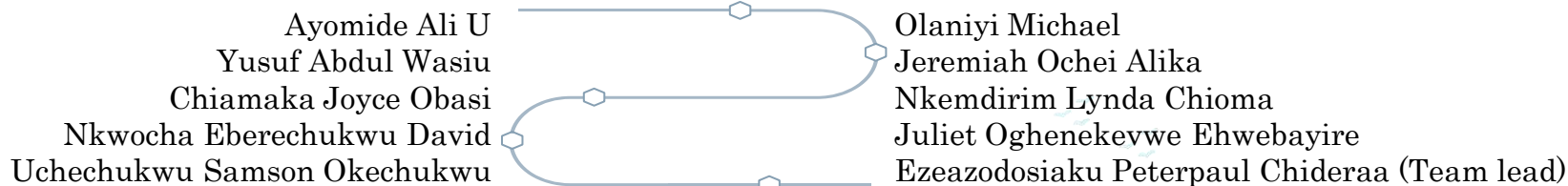


EMPLOYEE ATTRITION PREDICTION USING MACHINE LEARNING

A TechCrush Data Science Capstone Project (Cohort 7)

Presented by

Data Zest Group



Business Problem

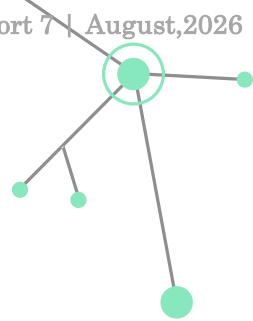
Employee turnover is a significant challenge for organisations because replacing experienced employees increases recruitment costs, reduces productivity, and disrupts team performance.

Project Objectives

- ❑ Identify factors associated with employee attrition.
- ❑ Explore employee behaviour using EDA.
- ❑ Build machine learning models to predict attrition.
- ❑ Compare model performance.
- ❑ Recommend strategies to improve employee retention.



Can machine learning help organisations identify employees who are likely to leave before they resign?



Dataset Overview

	Age	Attrition	BusinessTravel	DailyRate	Department	DistanceFromHome	Education	EducationField	EmployeeCount	EmployeeNumber	En
496	21	No	Travel_Rarely	1343	Sales	22	1	Technical Degree	1	669	
1271	21	Yes	Travel_Rarely	337	Sales	7	1	Marketing	1	1780	
1360	31	No	Travel_Rarely	471	Research & Development	4	3	Medical	1	1916	
752	36	Yes	Travel_Rarely	885	Research & Development	16	4	Life Sciences	1	1042	
70	43	No	Travel_Rarely	823	Research & Development	6	3	Medical	1	1866	

Dataset Summary

- ☐ 1,470 employee records
- ☐ 35 original features
- ☐ Binary classification problem

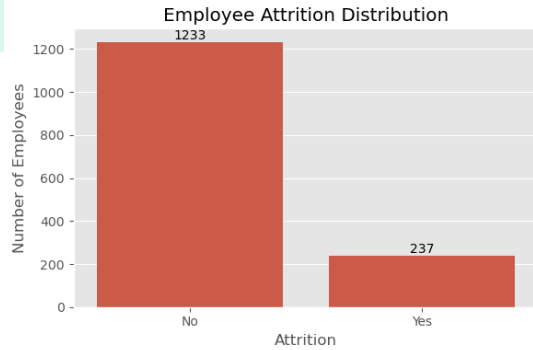
Target Variable:

- ☐ Attrition = Yes
- ☐ Attrition = No

Feature Categories

- ☐ Demographics
- ☐ Job Characteristics
- ☐ Compensation
- ☐ Performance
- ☐ Satisfaction
- ☐ Career History
- ☐ Workplace Factors

Exploratory Data Analysis



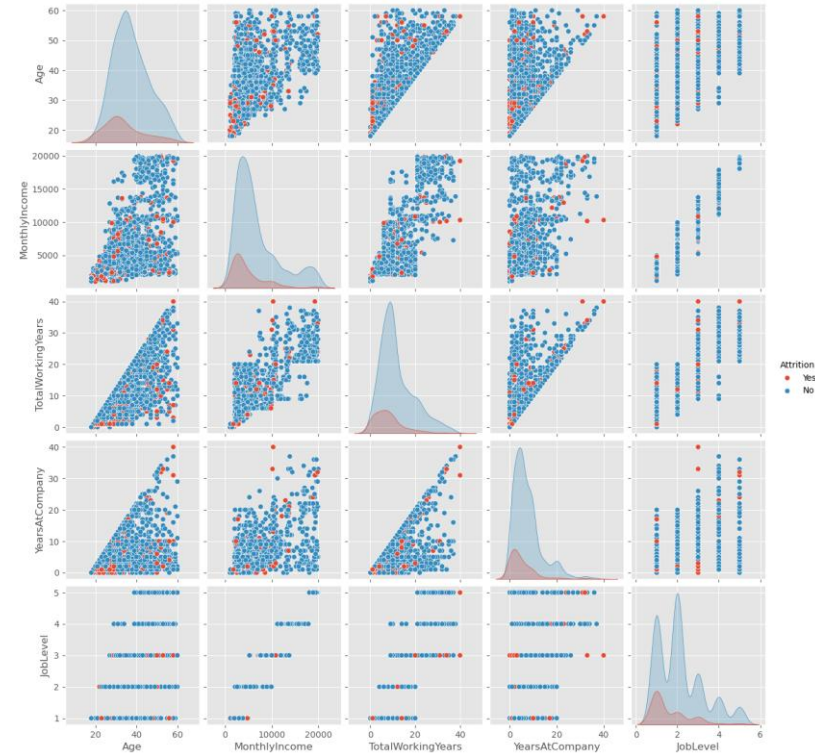
Attrition Distribution

Analyses Performed

- ☐ Data quality assessment
- ☐ Descriptive statistics
- ☐ Attrition distribution
- ☐ Pairplot analysis
- ☐ Statistical testing
- ☐ Chi-square analysis
- ☐ Cramér's V
- ☐ Point-biserial correlation

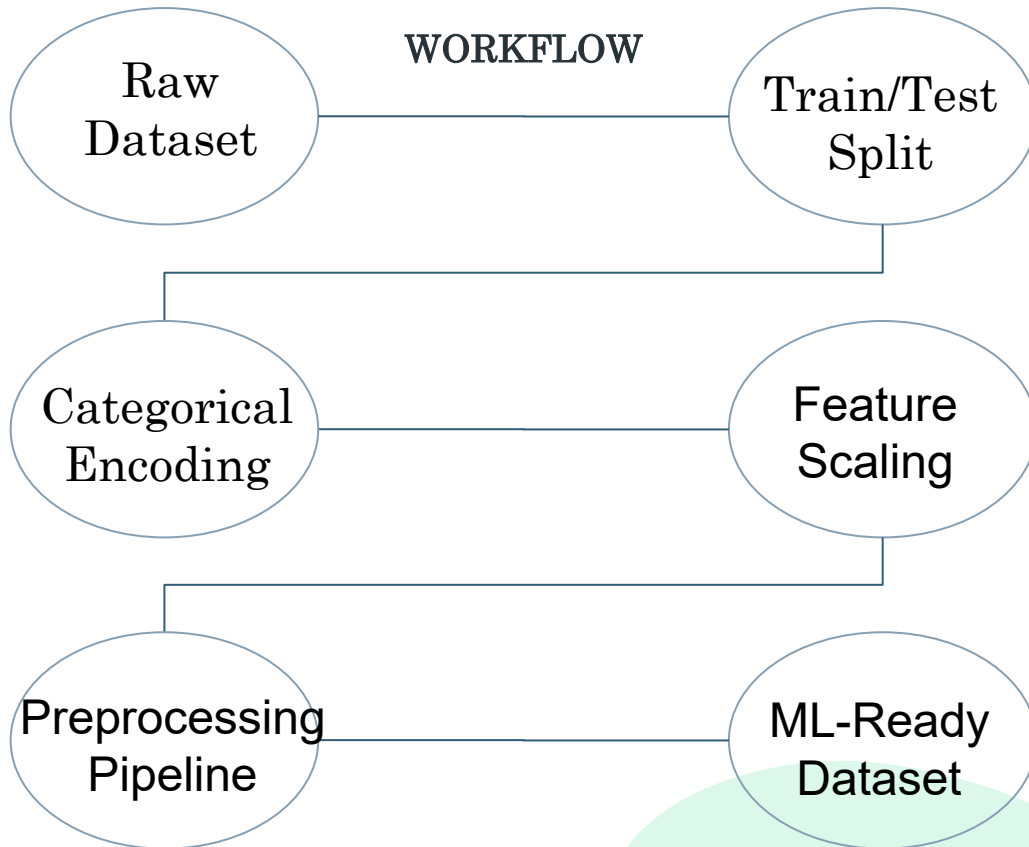
Key Findings

- ☐ Employees working overtime were more likely to leave.
- ☐ Career progression influenced attrition.
- ☐ Satisfaction-related variables showed meaningful relationships.
- ☐ Employee attrition is driven by multiple workplace factors rather than a single variable.



Pairplot Analysis

Feature Engineering And Preprocessing



Best Practices Applied

- ❑ Removed irrelevant variables
- ❑ Stratified train-test split
- ❑ Prevented data leakage
- ❑ Encoded categorical variables
- ❑ Standardised numerical variables
- ❑ Saved preprocessing pipeline using Joblib

Final Dataset

- ❑ Training: $1,176 \times 44$
- ❑ Testing: 294×44

Model Development

Three classification algorithms were developed and compared

Model

- ☐ Logistic Regression
- ☐ Support Vector Machine
- ☐ Gradient Boosting

Purpose

- Baseline statistical model
- Complex decision boundaries
- Ensemble learning

Evaluation Metrics

- ☐ Accuracy
- ☐ Precision
- ☐ Recall
- ☐ F1-Score
- ☐ ROC-AUC



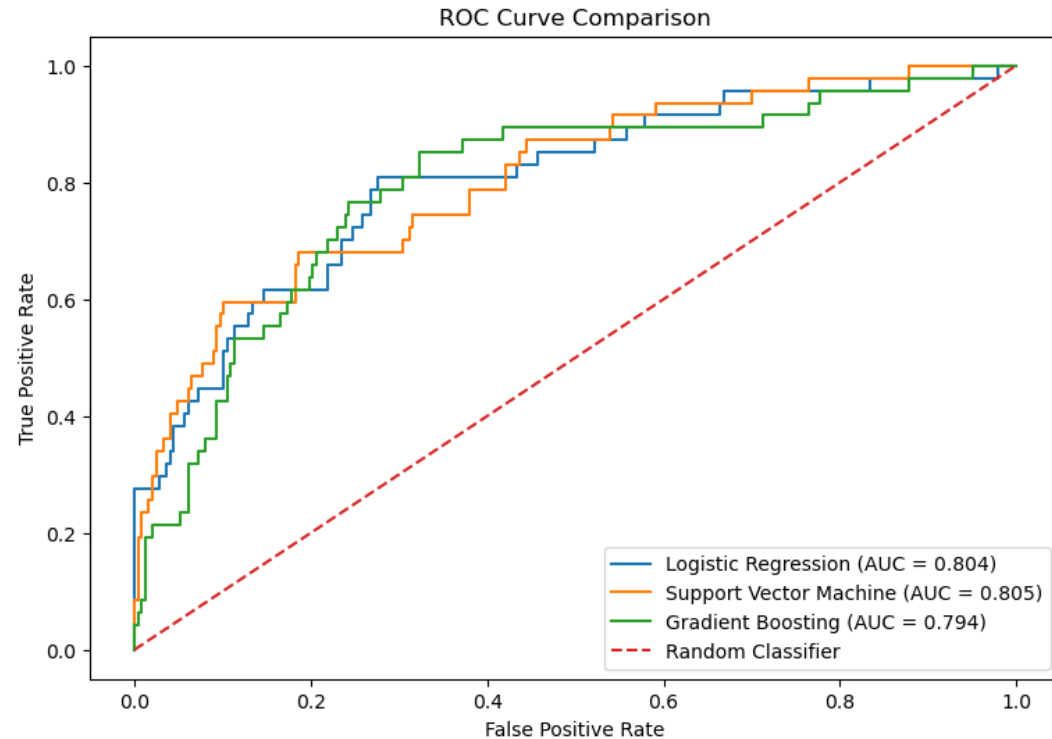
Model Evaluation

Model Performance

	Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
0	Logistic Regression	0.7551	0.3563	0.6596	0.4627	0.8035
1	Support Vector Machine	0.8163	0.4444	0.5957	0.5091	0.8054
2	Gradient Boosting	0.8503	0.5882	0.2128	0.3125	0.7941

Support Vector Machine (SVM) is the final Model since it has:

- ❑ Highest F1-score
- ❑ Highest ROC-AUC
- ❑ Best balance between Precision and Recall
- ❑ Better suited for identifying employees at risk of leaving



Feature Importance And Recommendations

```
feature_importance.head(15).round(4)
```

[16]:

	Feature	Importance Mean	Importance Std
43	cat_OverTime_Yes	0.1223	0.0218
11	num_NumCompaniesWorked	0.0353	0.0202
15	num_StockOptionLevel	0.0285	0.0327
21	num_YearsSinceLastPromotion	0.0225	0.0174
2	num_DistanceFromHome	0.0200	0.0225
5	num_HourlyRate	0.0174	0.0092
16	num_TotalWorkingYears	0.0110	0.0367
4	num_EnvironmentSatisfaction	0.0096	0.0105
18	num_WorkLifeBalance	0.0092	0.0190
8	num_JobSatisfaction	0.0084	0.0205
40	cat_JobRole_Sales Representative	0.0072	0.0047
17	num_TrainingTimesLastYear	0.0069	0.0211
34	cat_JobRole_Laboratory Technician	0.0063	0.0121
9	num_MonthlyIncome	0.0060	0.0247
22	num_YearsWithCurrManager	0.0059	0.0223

Key Predictors of Attrition

Top influential features:

- ☐ OverTime
- ☐ Number of Companies Worked
- ☐ Stock Option Level
- ☐ Years Since Last Promotion

Business Recommendations

- ☐ Monitor employee overtime.
- ☐ Improve career progression opportunities.
- ☐ Enhance employee satisfaction.
- ☐ Promote work-life balance.
- ☐ Use predictive analytics to support HR decisions.

Conclusion

- ❑ Successfully cleaned, explored, and prepared the IBM HR Analytics dataset.
- ❑ Developed and compared three machine learning classification models.
- ❑ **Support Vector Machine (SVM)** delivered the best overall balance of predictive performance.
- ❑ Identified key drivers of employee attrition, including:
 - ✓ *OverTime*
 - ✓ *Career progression*
 - ✓ *Work-life balance*
 - ✓ *Employee satisfaction*

This solution enables organisations to identify employees at risk of leaving and supports **proactive, data-driven retention strategies**, helping reduce turnover costs and improve workforce stability.

By transforming employee data into actionable insights, organisations can make smarter HR decisions, improve employee retention, and build a more engaged workforce."

Thank You

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